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Bereken de volgende bepaalde integralen door partiële integratie

$$\int_0^{\frac{\pi}{2}} x^2 \sin x \cos^3 x \cdot dx$$

onbepaalde integraal:

$$\begin{cases} u = x^2 \\ dv = \sin x \cos^3 x \cdot dx \end{cases} \Rightarrow \begin{cases} du = 2x \cdot dx \\ v = -\frac{1}{4} \cos^4 x \end{cases}$$

$$I = -\frac{1}{4}x^2 \cos^4 x + \frac{1}{2} \int x \cos^4 x \cdot dx$$

Dubbel toepassen van de halveringsformule geeft:

$$\begin{aligned} \int x \cos^4 x \cdot dx &= \int x \left(\frac{3}{8} + \frac{1}{2} \cos(2x) + \frac{1}{8} \cos(4x) \right) \cdot dx \\ &= \frac{3}{16}x^2 + \frac{1}{2} \int x \cos(2x) \cdot dx + \frac{1}{8} \int x \cos(4x) \cdot dx \end{aligned}$$

De termen $\int x \cos(ax) \cdot dx$:

$$\int x \cos(ax) \cdot dx = \frac{x}{a} \sin(ax) + \frac{1}{a^2} \cos(ax)$$

$$I = -\frac{1}{4}x^2 \cos^4 x + \frac{3}{32}x^2 + \frac{x}{8} \sin(2x) + \frac{1}{16} \cos(2x) + \frac{x}{64} \sin(4x) + \frac{1}{256} \cos(4x) + c$$

De bepaalde integraal:

$$I = \left[-\frac{1}{4}x^2 \cos^4 x + \frac{3}{32}x^2 + \frac{x}{8} \sin(2x) + \frac{1}{16} \cos(2x) + \frac{x}{64} \sin(4x) + \frac{1}{256} \cos(4x) \right]_0^{\frac{\pi}{2}}$$

bovengrens: $\frac{\pi}{2}$:

$$\left(0 + \frac{3\pi^2}{128} + 0 - \frac{1}{16} + 0 + \frac{1}{256} \right) = \frac{3\pi^2}{128} - \frac{15}{256}$$

ondergrens: 0:

$$\left(0 + 0 + 0 + \frac{1}{16} + 0 + \frac{1}{256} \right) = \frac{17}{256}$$

$$I = \left(\frac{3\pi^2}{128} - \frac{15}{256} \right) - \frac{17}{256} = \frac{3\pi^2}{128} - \frac{32}{256} = \frac{3\pi^2}{128} - \frac{1}{8}$$

References